

# UCab — Full-Stack MERN Cab Booking Platform

Solo Developer / Team Member: Shaik Sumiya Zainab | Project ID: N/A (Solo Track)

## Phase 8: Project Demonstration — Solo Developer Execution & Ownership Audit

Project Name: Cab Booking (UCab)

Project ID: N/A (Solo Track Submission)

Team Member / Solo Developer: Shaik Sumiya Zainab

### 1. Solo Track Ownership Declaration

This project (UCab) was designed, engineered, tested, and documented completely by a single developer (Shaik Sumiya Zainab). Rather than dividing workload across multiple team members, full-stack accountability was maintained across every layer of the application stack.

### 2. Complete Lifecycle Ownership Matrix

Project Domain / Discipline Specific Responsibilities Handled Key Deliverables & Artifacts Produced

**1. UI/UX & Design Systems** Conducted user empathy mapping for riders and dispatchers; designed custom glassmorphism CSS design system (index.css) with curated typography and color tokens. Empathy\_Map.md, Customer\_Journey\_Map.md, index.css, App.css.

**2. Frontend Engineering (React + Vite)** Built 11 distinct user and admin page components; constructed role-protected route wrappers (ProtectedRoutes.jsx); managed global session via AuthContext.jsx. Home.jsx, BookCab.jsx, UserHome.jsx, ReceiptModal.jsx, AdminCars.jsx, AuthContext.jsx.

**3. Backend API Development (Node + Express)** Architected RESTful endpoints (/api/users, /api/cars, /api/bookings, /api/admin); implemented bcrypt password hashing and JWT Bearer token authentication. server.js, authMiddleware.js, userController.js, carController.js, bookingController.js.

**4. Database Engineering & ODM** Designed relational Mongoose schemas (User, Car, Booking, Admin); engineered the Zero-Config Dual-Engine Storage Adapter (store.js) for universal grading resilience. User.js, Car.js, Booking.js, config.js, store.js, data.json.

**5. Quality Assurance & Testing** Executed comprehensive end-to-end testing across login flows, promo code discounting (UCAB20), and simulated offline database failover (data.json). Test\_Cases\_and\_Performance.md, Performance\_Testing.md.

**6. Documentation & Template Sync** Formatted and produced all exhaustive Phase 1 through Phase 8 markdown reports adhering strictly to the AI-ML and GEN-AI Track Project Template. All .md deliverable reports inside folders 1 through 8.

### 3. Solo Presentation & Defense Roles

During evaluation and project viva, the solo developer will execute all demonstration roles seamlessly:

- Architect Role: Explaining the rationale behind the dual-engine storage adapter (store.js) and system DFDs.
- Rider Role: Demonstrating live cab selection, promo discounting (UCAB20), and QR invoice generation (ReceiptModal.jsx).
- Dispatcher Role: Demonstrating real-time fleet inventory CRUD (AddCar.jsx) and dispatch status synchronization (AdminBookings.jsx).